



UNIVERSIDAD DE ANTIOQUIA
1803

Effective two-component and two-mediator dark matter

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In collaboration with

Kimy Agudelo, Andrés Rivera and David Suárez (UdeA)



Baryon and Lepton Nonconserving Processes

Steven Weinberg (Harvard U.) (1979)

Published in: *Phys.Rev.Lett.* 43 (1979) 1566-1570

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#1 Systematic Study of One-Loop Dirac Neutrino Masses and Viable Dark Matter Candidates

Chang-Yuan Yao (Hefei, CUST), Gui-Jun Ding (Hefei, CUST) (Jul 31, 2017)

Published in: *Phys.Rev.D* 96 (2017) 9, 095004, *Phys.Rev.D* 98 (2018) 3, 039901 (erratum) • e-Print: [1707.09786](#) [hep-ph]

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Name	$SU(2)_L$	$U(1)_Y$
L	2	$-1/2$
H	2	$1/2$

$$\mathcal{L}_M = L \cdot H L \cdot H + \text{h.c}$$

Dark matter (radiative) realization $\rightarrow Z_2$

$$\mathcal{L}_D = (\nu_R)^\dagger L \cdot H + \text{h.c}$$

$$\overline{M_{N^c R^c R}} \rightarrow U(1)$$

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Scotogenic one-loop realizations

Majorana

Dirac

Systematic study of the d=5 Weinberg operator at one-loop order

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Florian Bonnet (Wurzburg U.), Martin Hirsch (Valencia U., IFIC), Toshihiko Ota (Munich, Max Planck Inst.), Walter Winter (Wurzburg U.) (Apr, 2012)

Published in: *JHEP* 07 (2012) 153 • e-Print: [1204.5862](#) [hep-ph]

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Models with radiative neutrino masses and viable dark matter candidates

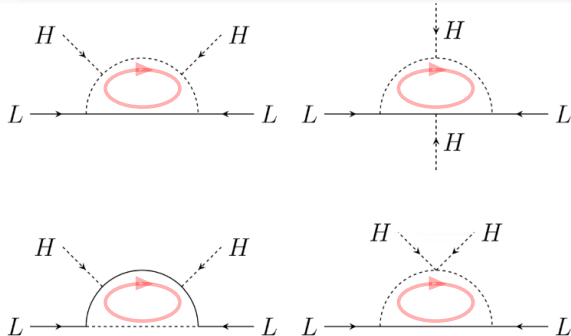
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Diego Restrepo (Antioquia U.), Oscar Zapata (Antioquia U.), Carlos E. Yaguna (Munster U., ITP) (Aug 16, 2013)

Published in: *JHEP* 11 (2013) 011 • e-Print: [1308.3655](#) [hep-ph]

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Pathways to Naturally Small Dirac Neutrino Masses

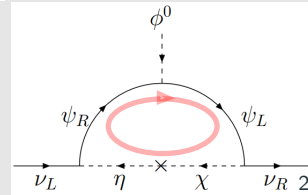
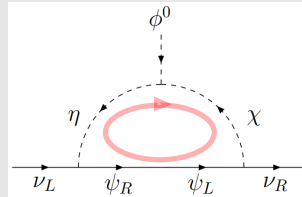
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Ernest Ma (UC, Riverside), Oleg Popov (UC, Riverside) (Sep 8, 2016)

Published in: *Phys.Lett.B* 764 (2017) 142-144 • e-Print: [1609.02538](#) [hep-ph]

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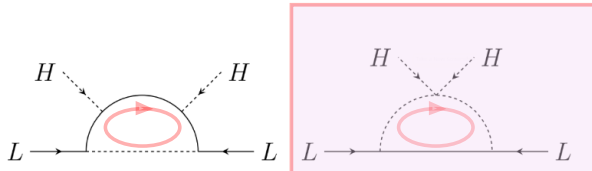
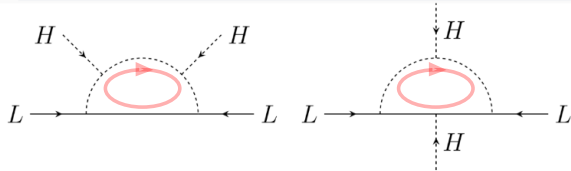
Models with radiative neutrino masses and viable dark matter candidates

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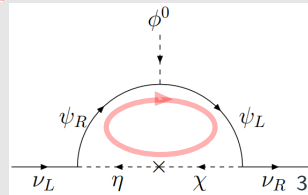
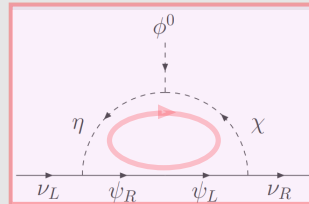
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Radiative seesaw mechanism at weak scale

Zhi-jian Tao (Beijing, Inst. High Energy Phys.) (Mar, 1996)

Published in: *Phys.Rev.D* 54 (1996) 5693-5697 • e-Print: [hep-ph/9603309](https://arxiv.org/abs/hep-ph/9603309) [hep-ph]

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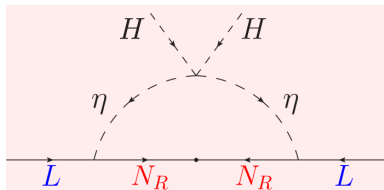
Verifiable radiative seesaw mechanism of neutrino mass and dark matter

Ernest Ma (UC, Riverside) (Jan, 2006)

Published in: *Phys.Rev.D* 73 (2006) 077301 • e-Print: [hep-ph/0601225](https://arxiv.org/abs/hep-ph/0601225) [hep-ph]

pdf DOI cite claim reference search 1,539 citations

Name	$SU(2)_L$	$U(1)_Y$	Z_2
N_R	1	0	—
η	2	1/2	—



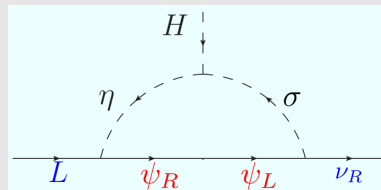
Radiative Neutrino Mass, Dark Matter and Leptogenesis

Pei-Hong Gu (ICTP, Trieste), Utpal Sarkar (Ahmedabad, Phys. Res. Lab) (Dec, 2007)

Published in: *Phys.Rev.D* 77 (2008) 105031 • e-Print: [0712.2933](https://arxiv.org/abs/0712.2933) [hep-ph]

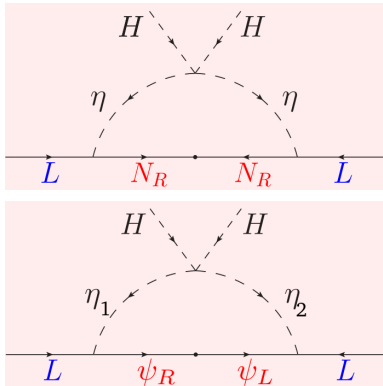
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Name	$SU(2)_L$	$U(1)_Y$	Z_2	$U(1)_L$
ν_R	1	0	+	-1
ψ_R	1	0	—	-1
ψ_L	1	0	—	-1
η	2	1/2	—	0
σ	1	0	—	0



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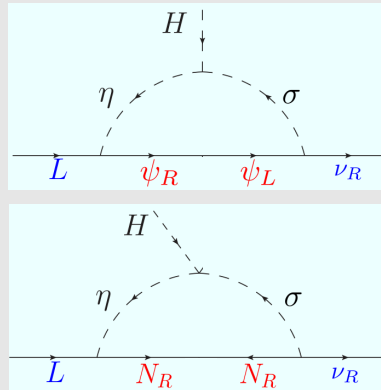
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New Scotogenic Model of Neutrino Mass with $U(1)_D$ Gauge Interaction

Ernest Ma (UC, Riverside), Ivica Picek (Zagreb U., Phys. Dept.), Branimir Radovčić (Zagreb U., Phys. Dept.) (Aug 24, 2013)

Published in: *Phys.Lett.B* 726 (2013) 744-746 • e-Print: [1308.5313](#) [hep-ph]
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Dirac neutrino mass generation from a Majorana messenger

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U. and IIP, Brazil), Óscar Zapata (Antioquia U.) (Sep 20, 2019)

Published in: *Phys.Rev.D* 101 (2020) 3, 035004 • e-Print: [1909.09574](#) [hep-ph]
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